WORKED SOLUTION

STEP I 1995, Question 1

Pure • Polynomial Inequalities

STEP I 1995, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q1

The key idea

Part (i) is a cubic that factorises by grouping, and once it is a product of three linear factors its sign can be read off from a number line. Part (ii) is the same cubic in disguise: every term has total degree 3, so dividing by y^3 turns it into the cubic of part (i) in the variable x/y. The same factorisation then gives three lines through the origin, and part (iii) is a sign chart drawn around the origin instead of along a line.

Solution

Part (i)

Grouping the terms in pairs,

$$\begin{aligned} x^3 - 4x^2 - x + 4 &= x^2(x-4) - (x-4) \\ &= (x-4)(x^2-1) \\ &= (x+1)(x-1)(x-4). \end{aligned}$$

The roots are -1, 1 and 4. Each factor is linear and changes sign only at its own root, so the product changes sign at each of the three roots. For large positive x all three factors are positive, so the sign pattern, reading from the right, is

x	$x < -1$	$-1 < x < 1$	$1 < x < 4$	$x > 4$
sign	$-$	$+$	$-$	$+$

Including the roots, where the cubic is zero, the inequality holds exactly when

$$-1 \leq x \leq 1 \quad \text{or} \quad x \geq 4.$$

Part (ii)

Write $f(x,y) = x^3 - 4x^2y - xy^2 + 4y^3$. It is part (i)'s cubic with each term made up to degree 3 by powers of y, and the same grouping works:

$$\begin{aligned} f(x,y) &= x^2(x-4y) - y^2(x-4y) \\ &= (x-4y)(x-y)(x+y). \end{aligned}$$

A product is zero exactly when one of its factors is zero, so $f(x,y) = 0$ on the three lines

$$x = 4y \text{ (i.e. } y = \tfrac{1}{4}x), \qquad y = x, \qquad y = -x,$$

all of which pass through the origin.

? **Ask yourself** Why should the answer to (ii) have anything to do with (i)?

For $y \neq 0$, put $t = x/y$. Then $f(x,y) = y^3\,(t^3 - 4t^2 - t + 4)$, which is y^3 times the cubic of part (i). That cubic vanishes at $t = -1, 1, 4$, that is, on $x = -y$, $x = y$ and $x = 4y$. (When $y = 0$, $f = x^3$ vanishes only at the origin, which is already on all three lines.)

Part (iii)

The three lines cut the plane into six sectors meeting at the origin. Measured anticlockwise from the positive x-axis, the six half-lines point in the directions

$$\arctan\tfrac{1}{4} \approx 14^\circ, \quad 45^\circ, \quad 135^\circ, \quad 194^\circ, \quad 225^\circ, \quad 315^\circ.$$

Inside a sector no factor of f is zero, so f has a constant sign there. Crossing one of the lines changes the sign of exactly one factor (the three lines are distinct, and away from the origin we cross only one at a time), so the sign of f alternates from one sector to the next.

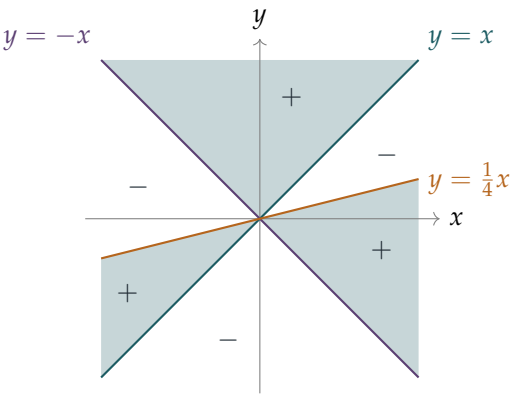
To fix the signs, take $(1,0)$: $f(1,0) = 1 > 0$, and $(1,0)$ lies in the sector between 315° and 14°. Going round, the signs are therefore

sector (degrees)	$-45, 14$	$14, 45$	$45, 135$	$135, 194$	$194, 225$	$225, 315$
sign of f	$+$	$-$	$+$	$-$	$+$	$-$

As a spot check, $f(0,1) = 4 > 0$ in the third column and $f(-3,-1) = 8 > 0$ in the fifth.

So $f(x,y) \geq 0$ on the three shaded sectors below, together with all three lines (where $f = 0$):

- the sector containing the positive x-axis, between $y = -x$ (below) and $y = \frac{1}{4}x$ (above), for $x > 0$;
- the sector containing the positive y-axis, between $y = x$ and $y = -x$, for $y > 0$;
- the thin sector in the third quadrant between $y = x$ and $y = \frac{1}{4}x$.



Ask yourself Why is the picture unchanged by a half-turn about the origin, with the shading swapped?

Every term of f has odd degree 3, so $f(-x,-y) = -f(x,y)$. Rotating a point through $180°$ about the origin flips the sign of f, so each shaded sector sits opposite an unshaded one. That is a quick check on any sketch: shaded and unshaded sectors must come in opposite pairs, and they do.

Checks

- On the line $y = 1$, $f(x,1)$ is exactly the cubic of part (i), so the shaded part of $y = 1$ should be $-1 \le x \le 1$ and $x \ge 4$. In the sketch, $y = 1$ is shaded between $y = -x$ and $y = x$ (from $x = -1$ to $x = 1$), and it re-enters a shaded sector where it crosses $y = \frac{1}{4}x$, at $x = 4$.
- On the x-axis $f = x^3$, which is non-negative exactly for $x \ge 0$: the positive x-axis is shaded and the negative one is not.
- Numerically, f was evaluated at 2×10^5 random points, and its sign agreed with the six-sector chart every time; for instance $f(1,0) = 1$, $f(0,-1) = -4$, $f(-3,-1) = 8$.

Common slips

- Dividing by y^3 and keeping the inequality sign for $y < 0$. Since $y^3 < 0$ there, the condition becomes $t^3 - 4t^2 - t + 4 \le 0$, which is why the lower half-plane looks different from the upper.
- Shading using part (i)'s answer for $y > 0$ only and then forgetting $y < 0$ altogether.
- Leaving out the lines themselves: the inequality is non-strict, so they belong to the region.

Beyond STEP

A polynomial whose terms all have the same degree is called *homogeneous*. Its zero set is a union of lines through the origin, and its sign depends only on direction. Homogeneous polynomials are the starting point of projective geometry, where a point is a whole line through the origin, and they appear again when you classify conics and quadratic forms in a linear algebra course. Strang's *Introduction to Linear Algebra* covers quadratic forms, and Siklos's *Advanced Problems in Mathematics* has more STEP questions in this style.

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